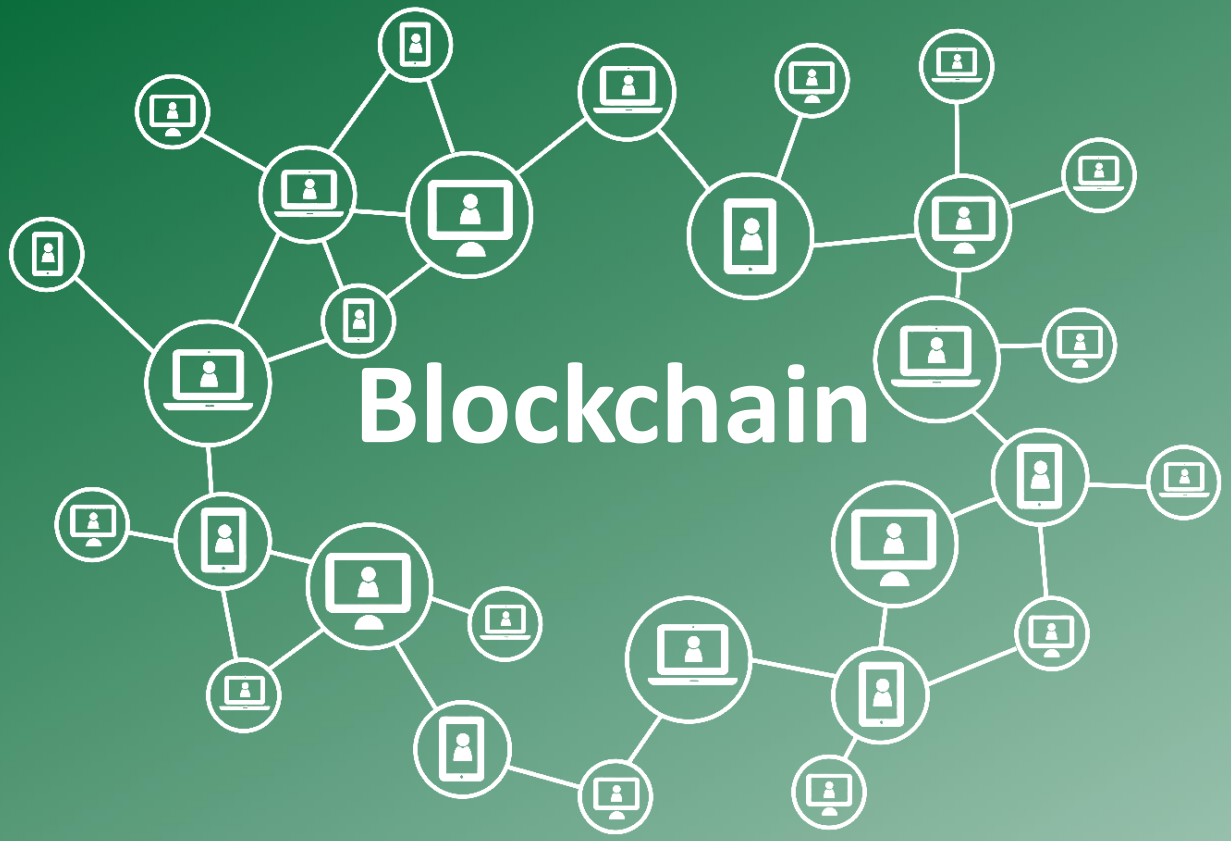


Blockchain and Cryptocurrency for Dummies



Blockchain Overview

Blockchain Timeline

October 2008:
Bitcoin whitepaper by the nom-de-plume Satoshi Nakamoto is published.

May 2010:
First Bitcoin purchase: BTC 10k for a $25 pizza. Today BTC 10k is worth $10m! Bitcoin is known as the first use case of Blockchain technology.

June 2014:
LHVpank starts research on Blockchain and its digital security with their app "Cuber Wallet".

July 2014:
Ethereum Project – a Blockchain platform with the ability to build decentralized applications – is funded by a crowd sale.

September 2015:
Major financial companies form R3 – a consortium of over 40 institutions committed to exploring and implementing Blockchain technology.

September 2015:
Visa, Citi, Nasdaq, Capital One and Fiserv invest $30m in the Blockchain startup Chain.com.

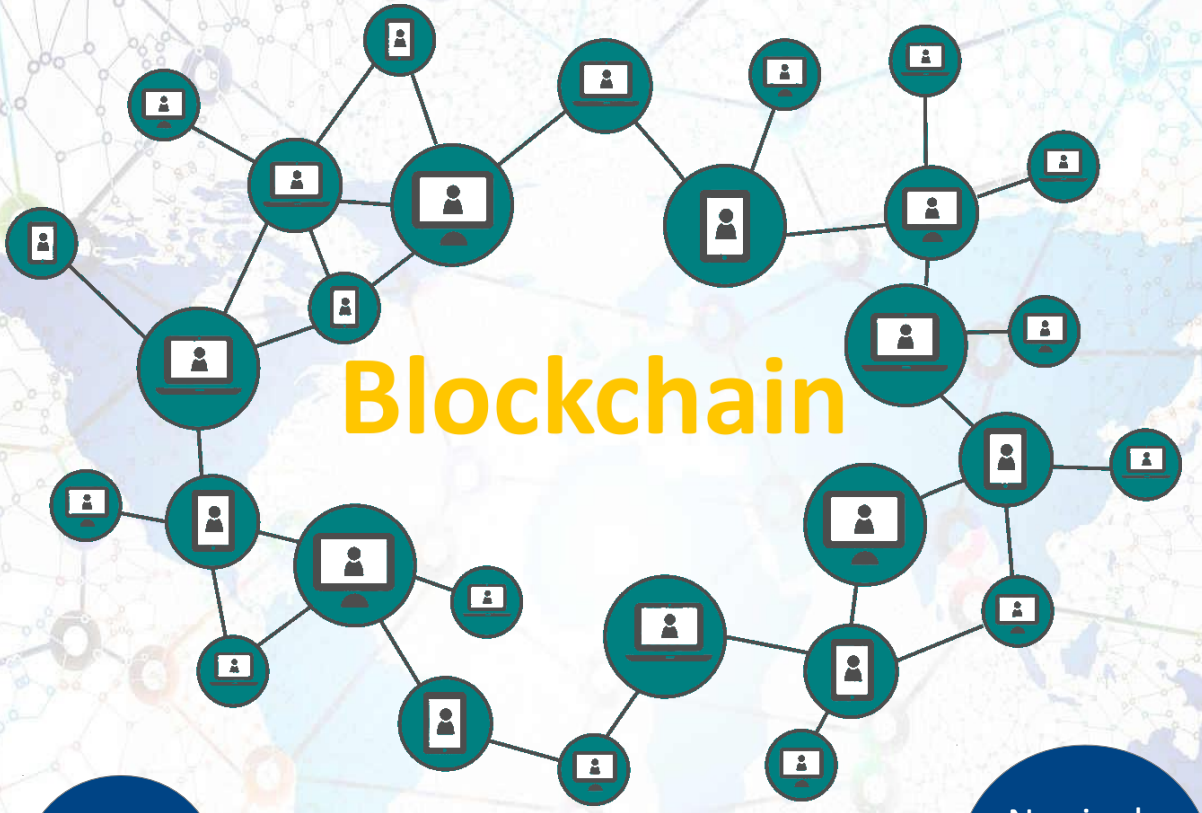
September 2016:
Over 40 financial service institutions have invested in a Blockchain or Bitcoin startup since 2014.

2018:
WEC estimates that 80% of all banks will initiate projects concerning distributed ledger technology – the underlying technology supporting Blockchain.

Souces: Oliver Wyman, 2016; MissQT Fintech, 2015; GEM Mann, 2016; CoinDesk – Blockchain Breakout? Will Keep Investors on the Brink, 2017; Deloitte – banking colloquium thought paper 2015

The Potential of Blockchain Technology, Pioneers Discover
https://www.slideshare.net/Pioneers_io/the-potential-of-blockchain-technology-72277655

Blockchain at a Glance



Shared

Ledger of transactions

Anyone can inspect the transactions

No single entity controls

Public vs Consortium vs Private Blockchain

	Public No Centralized Management	Consortium Multiple Organizations	Private Single Organization
Participants	Permissionless • Anonymous • Could be malicious	Permissioned • Identified • Trusted • Could misbehave	Permissioned • Identified • Trusted
Consensus Mechanisms	Proof of Work, Proof of Stake, etc.. • Large energy consumption • No finality • 51% attack	Voting or multi-party consensus algorithm • Lighter • Faster • Low energy consumption	Pre-approved participants • Lighter • Faster • Low energy consumption • Cheaper
Transaction Approval Freq.	Long Bitcoin: 10 min or more	Depends on number of nodes but faster than public blockchain	Short 100x msec

Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

https://bitcoin.org/bitcoin.pdf

Ethereum



- A blockchain platform that runs smart contracts
- Using Ether as a mean of payment (Cryptocurrency) but is listed in Cryptocurrency Exchange as “Ethereum”
- Ethereum allows different digital assets (tokens) which may be used in conjunction with Ether
- ERC-20 is one of the most significant token standards of all for Ethereum

Hyperledger



HYPERLEDGER
BLOCKCHAIN TECHNOLOGIES FOR BUSINESS

- The most well-known private blockchain frameworks intend for business
- Hosted by Linux Foundation
- Under the name Hyperledger, there are many frameworks and tools inside for different purposes



HYPERLEDGER SAWTOOTH



HYPERLEDGER IROHA



HYPERLEDGER INDY



HYPERLEDGER FABRIC



HYPERLEDGER BURROW



HYPERLEDGER CALIPER



HYPERLEDGER CELLO



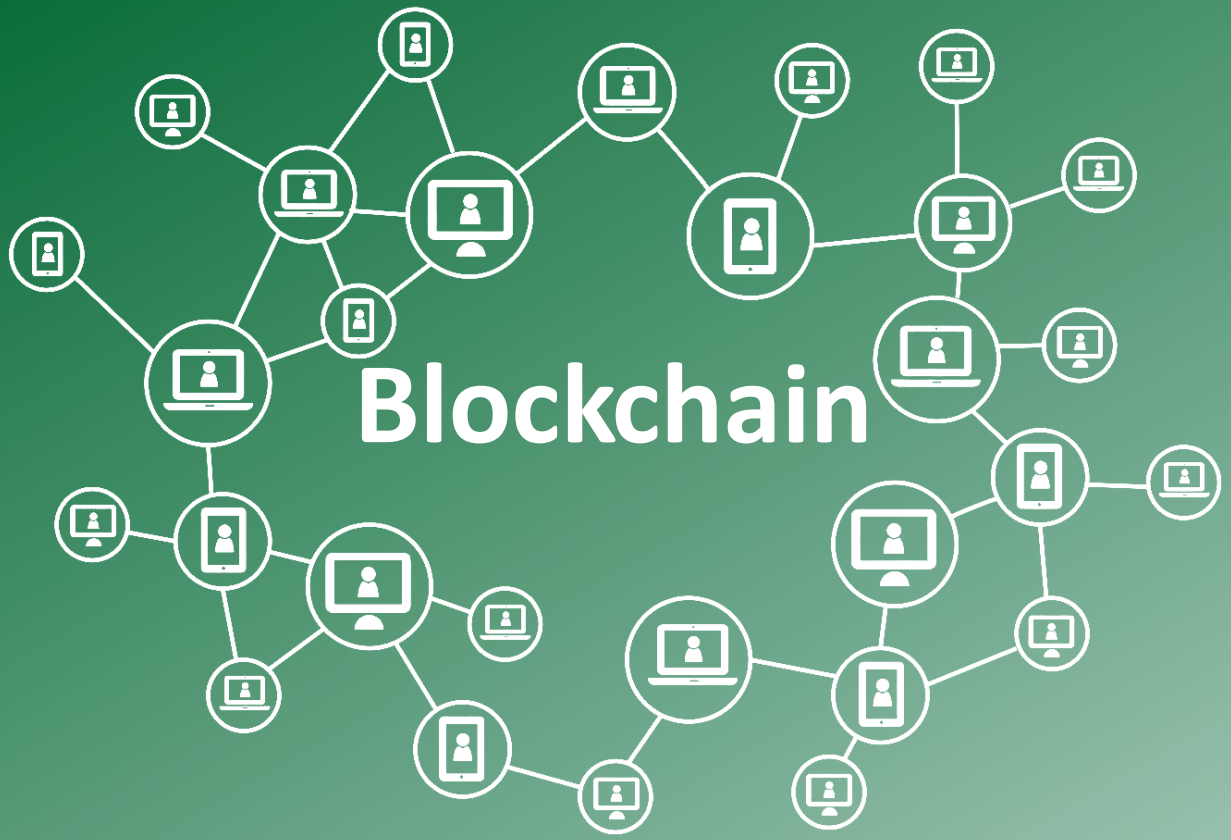
HYPERLEDGER COMPOSER



HYPERLEDGER QUILT



HYPERLEDGER EXPLORER



Basic Cryptography in Blockchain

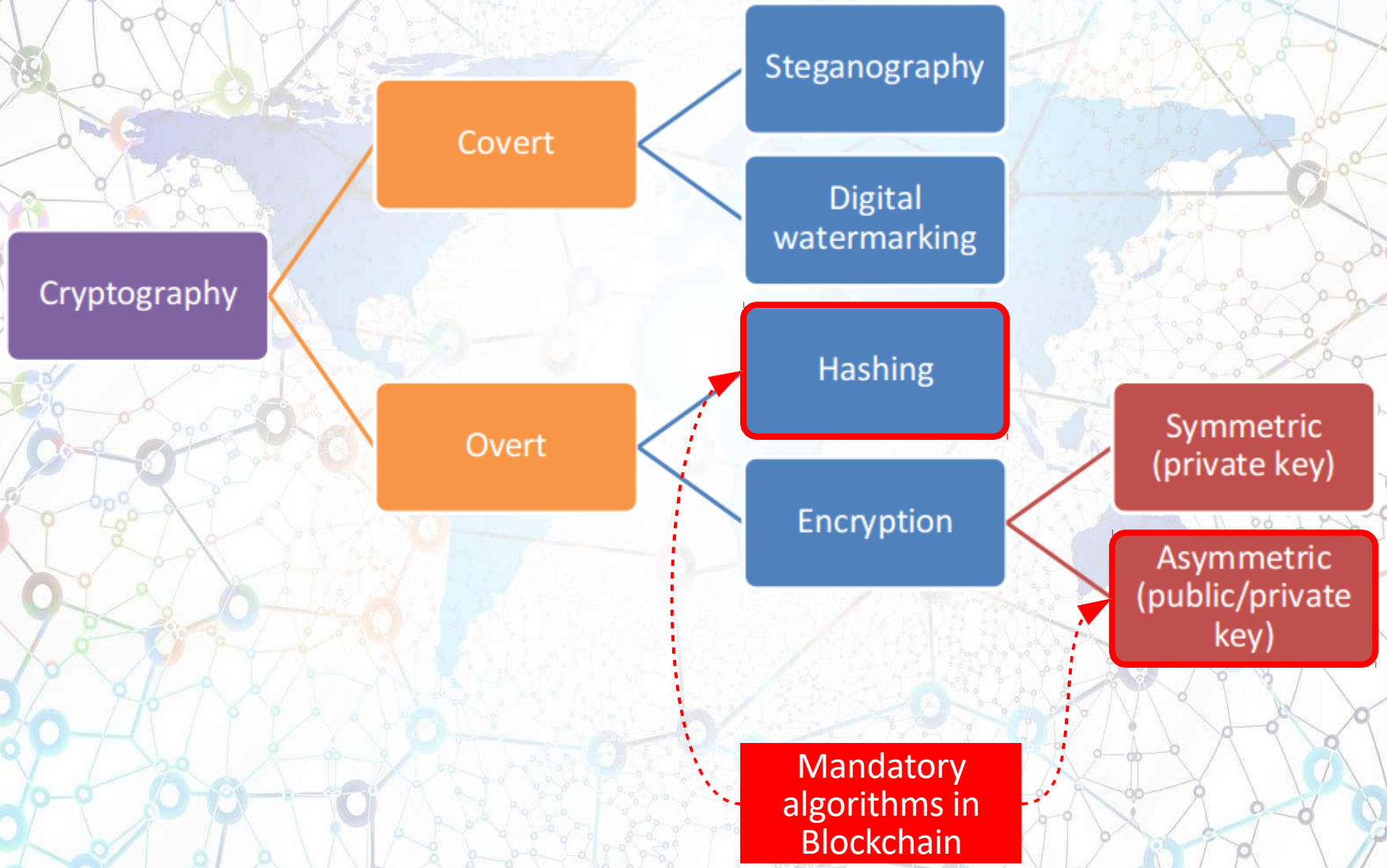
Cryptography Definitions & Concepts

- The process of converting ordinary plain text into unintelligible text and vice-versa
- Modern cryptography concerns with:
  - Confidentiality - Information cannot be understood by anyone
  - *Integrity - Information cannot be altered.*
  - *Non-repudiation - Sender cannot deny his/her intentions in the transmission of the information at a later stage*
  - Authentication - Sender and receiver can confirm each

Mandatory concerns in Blockchain

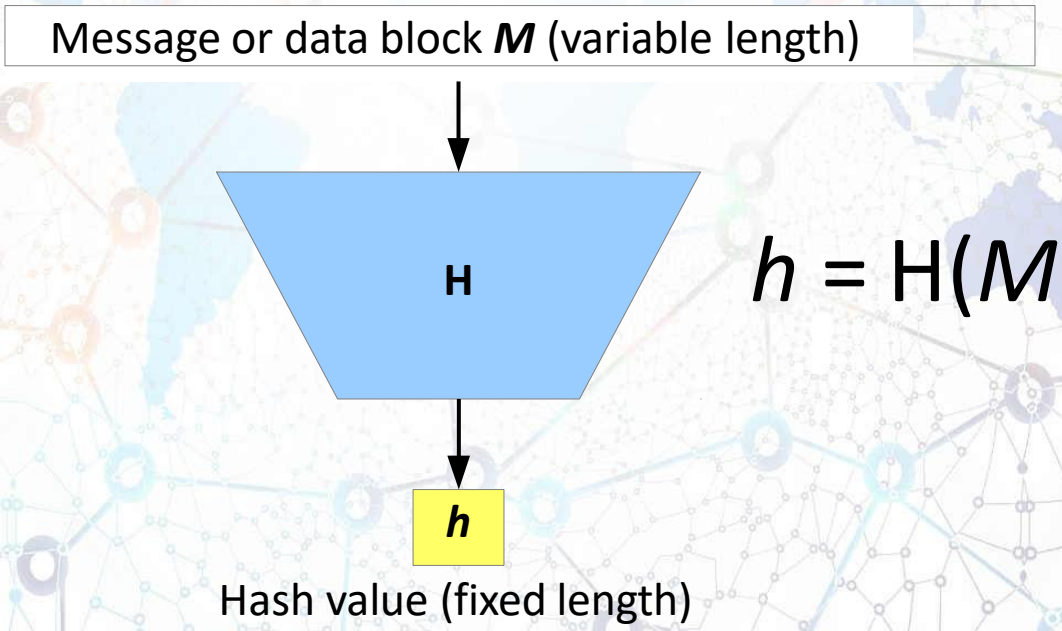
- Modern cryptography mainly based on mathematical theory and computer science practice

Types of Cryptography



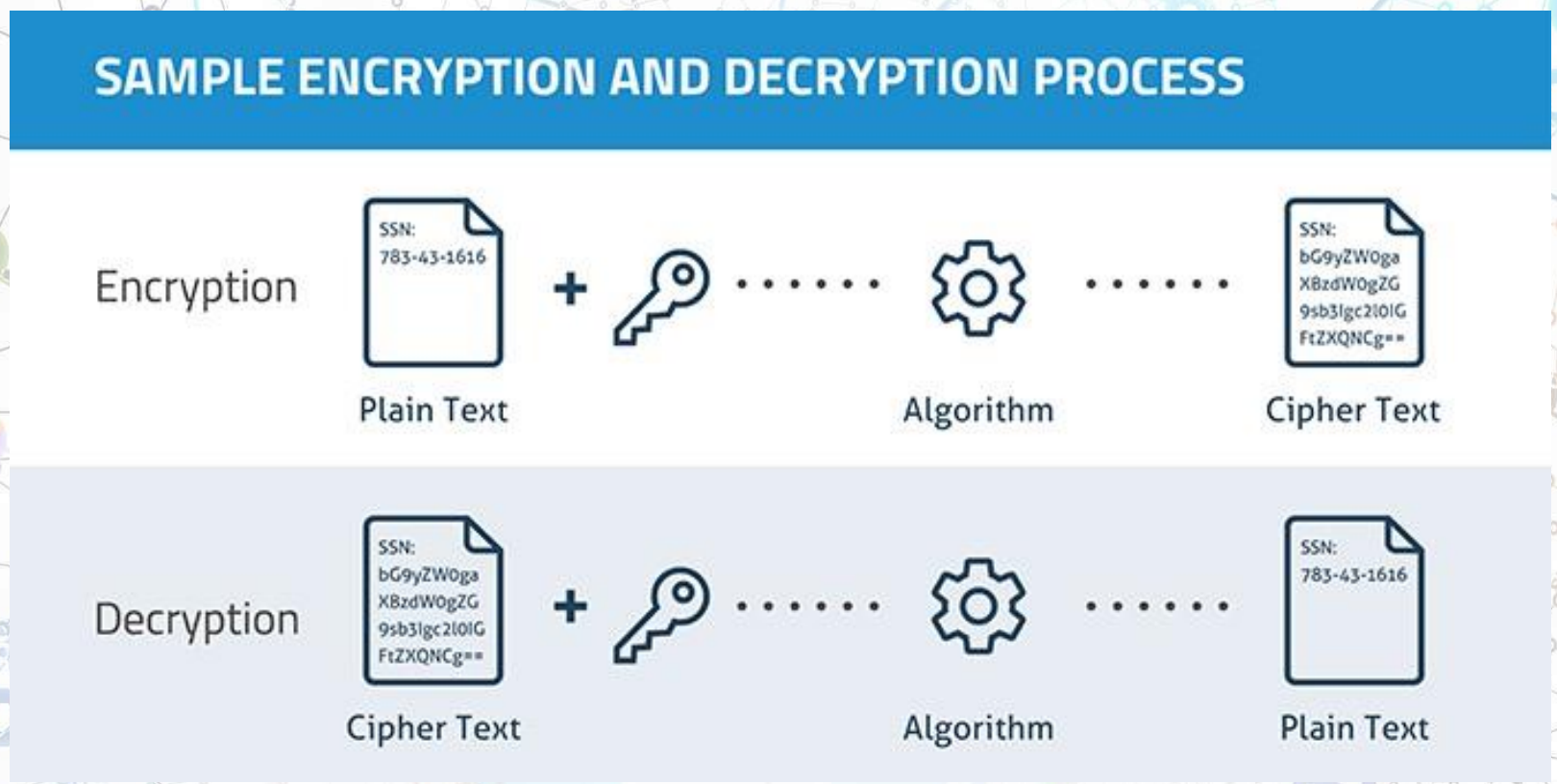
Hash Function

- Any function that can be used to map data of arbitrary size to data of a fixed size.
- The value returned by a hash function is called hash
- In the other hand, the hash is a fingerprint of the message
- Well-know hash functions: SHA-1, SHA-2



$h = \mathrm{H}(M)$

Encryption



Encryption is a method of transforming readable data, called plain text, into a form that appears to be random and unreadable, which is called cipher text. Plain text is in a form that can be understood either by a person (a document) or by a computer (executable code). Once it is transformed into cipher text, neither human nor machine can properly process it until it is decrypted.

Asymmetric Cryptography

- Aka “Public Key Cryptography”
- Two related keys (public and private key) are used
  - Public key may be freely distributed while its paired private key remains a secret
  - Either of the keys can be used to encrypt a message; the opposite key is used for decryption
- If a public key is authentic (belongs to the person or entity claimed) and that it has not been tampered with or replaced by a malicious third party, asymmetric encryption will deliver
  - Confidentiality
  - Integrity
  - Authenticity
  - Non-repudiation

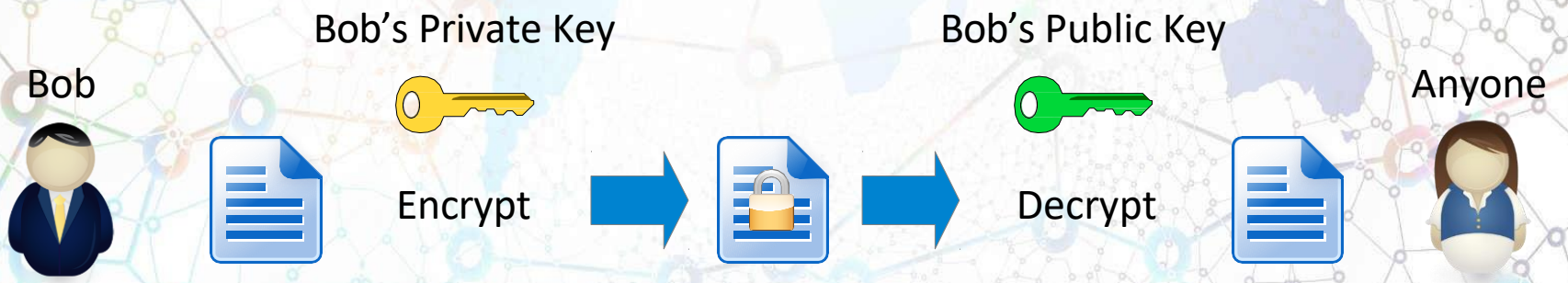
Two Usage of Asymmetric Encryption

Confidentiality assurance in asymmetric key cryptography



Nobody can read encrypted message except Bob.

Proof of origin assurance in asymmetric key cryptography



Everyone can read encrypted message with Bob's public key and know it is from Bob.

Digital Signature

Creating and verifying a digital signature



Calculate hashcode

1011...10101

Encrypt the hashcode with private key of the sender

1011...10101

Creation of a digitally signed document (sender)



Digitally signed document

1011...10101

Calculate hashcode

1011...10101

?

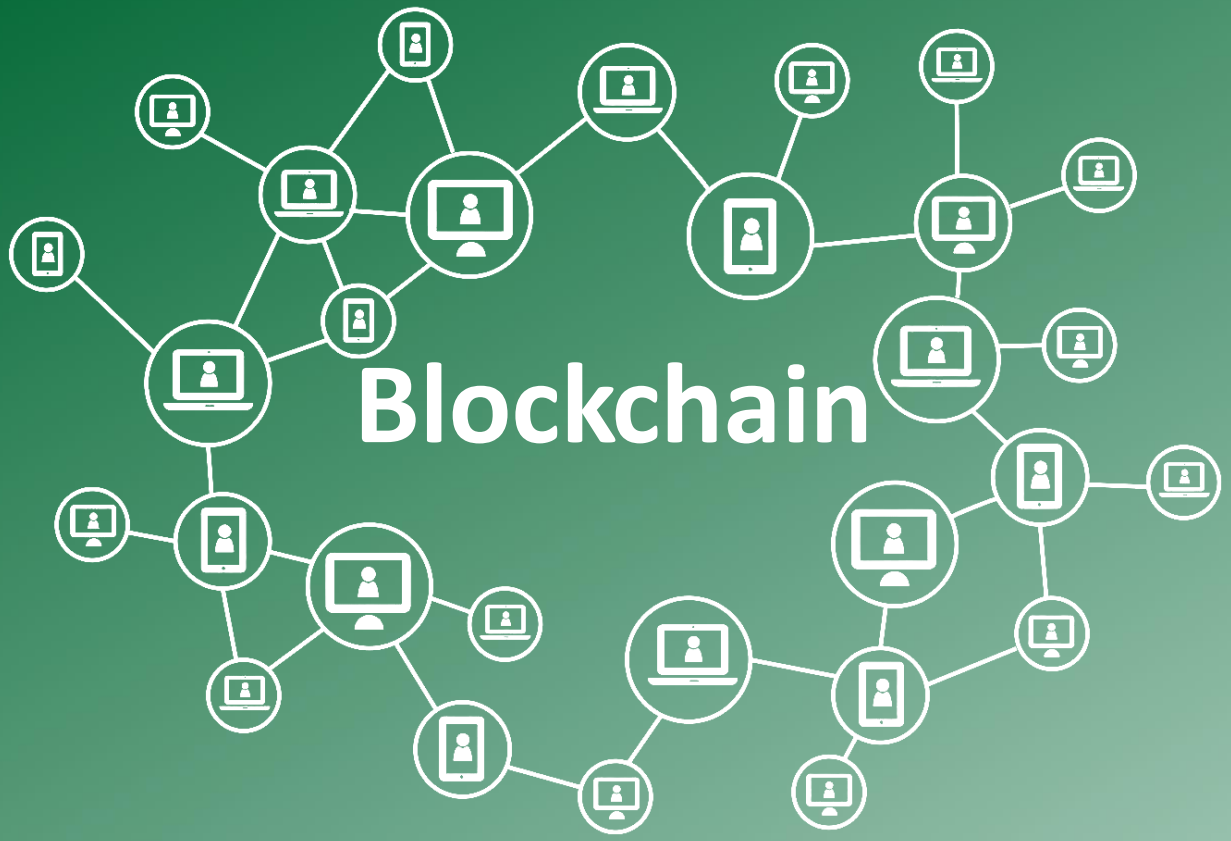
=

Decrypt the signature with the public key of the sender

1011...10101

Verifying the digital signature (receiver)

If the calculated hashcode does not match the result of the decrypted signature, either the document was changed after signing, or the signature was not generated with the private key of the alleged sender.



Blockchain Basics

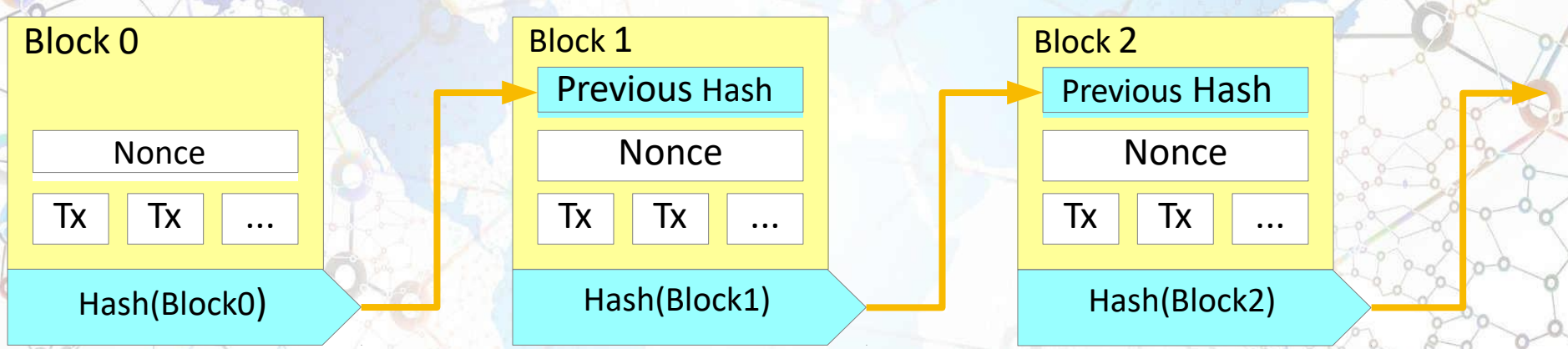
Blockchain Distinction

Blockchain technology must consist of these 3 properties

- A chain of blocks that metadata (or header) in each block contain the result from hash function of the previous block data except the Genesis block
- Decentralization with proven mechanism to ensure every node will obtain the same data during block creation process (consensus).
- Open and transparent execution

For public blockchains, balance benefit and incentive model must be declared

Chain of Hashes



Consensus Protocols

- Proof of Work (PoW)
  - Concept: Who can solve the problem first will get the incentive (mining) and choose which transactions to be in the next block.
  - Implementation: Bitcoin, Ethereum (current)
  - Attack Resistance: Attacker must have more than 50% of the whole network computing power
- Proof of Stake (PoS)
  - Concept: Who has the most of stakes (rich) can choose which transactions to be in the next block.
  - Implementation: Peercoin, Ethereum (planned)
  - Attack Resistance: Attacker must have more than 50% of the whole network stakes

Consensus Protocols (cont'd)

- Practical Byzantine Fault Tolerance (PBFT)
  - Concept: No mining, we vote a leader every time with the same rule and the leader will set parameters for the next block.
  - Implementation: Hyperledger
  - Attack Resistance: Attacker must have more than 1/3 of total nodes to stop block creation and 2/3 to manipulate transactions
- Hybrid
  - Concept: Each protocol has different strength, can we take the best of two or more protocols?
  - Implementation: Tendermint (PBFT+PoS)
  - Attack Resistance: Depends on which protocols

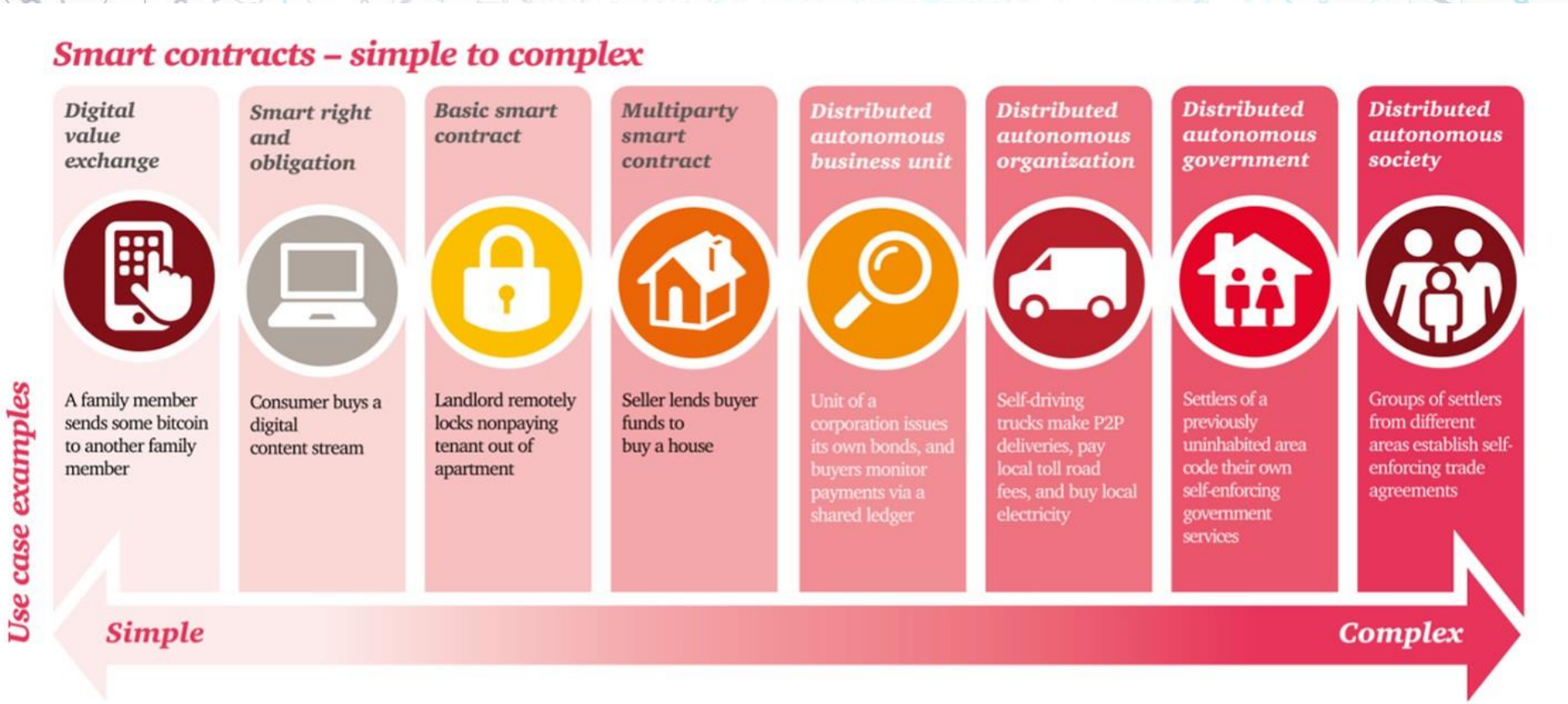
Open and Transparent Execution

- Designs and algorithms must be declared to public to verify
- Source code must be able to be audited in order to prove that declared designs and algorithms are implemented
- All transactions are traceable, and permanently stored in the blockchain network.

Smart Contract

- A computer code running on top of a blockchain containing a set of rules under which the parties to that smart contract agree to interact with each other.
- If and when the pre-defined rules are met, the smart contract will auto execute the transaction.
- The Ethereum project introduced the idea of decoupling the contract layer from the blockchain layer.
- A smart contract can only be as smart as the people coding taking into account all available information at the time of coding.

Smart Contract Examples



Source: PricewaterhouseCoopers
http://usblogs.pwc.com/emerging-technology/how-smart-contracts-automate-digital-business/

Typical Blockchain Technology Stack

SMART CONTRACTS
Relations



Define behavioural rulesets for all participants of the smart contract

APPLICATION LAYER

RECORD OF TRANSACTIONS
Assets



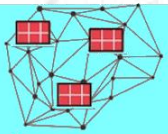
File containing all information since block 1 - tracking all asset movements.

CONSENSUS RULES
Governance



Defining game theoretical behavioural rulesets of all actors in the network

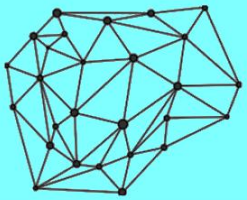
P2P NETWORK OF COMPUTERS
Physical Network



A network of all devices running the blockchain protocol, and keeping records of transactions

BLOCKCHAIN LAYER

TCP/IP
Infrastructure

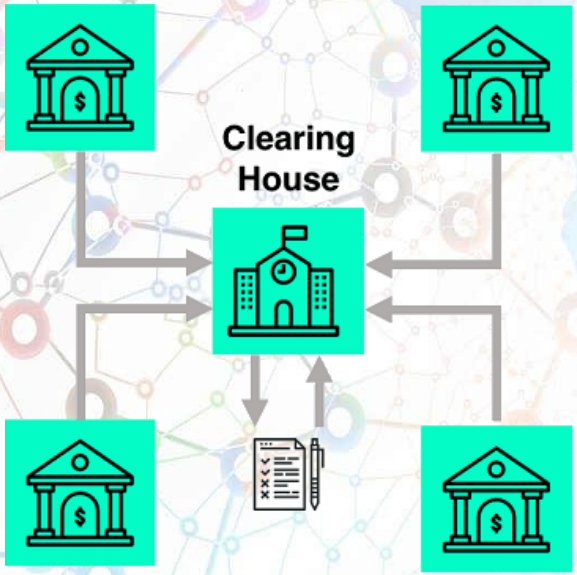


INTERNET LAYER

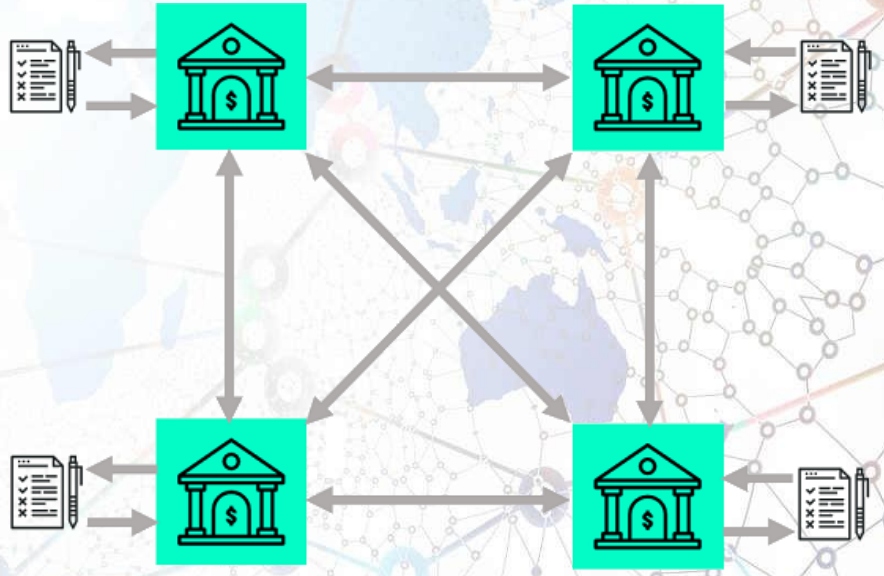
Blockchain: A Beginners Guide, BlockchainHub

Distributed Ledger Technology (DLT)

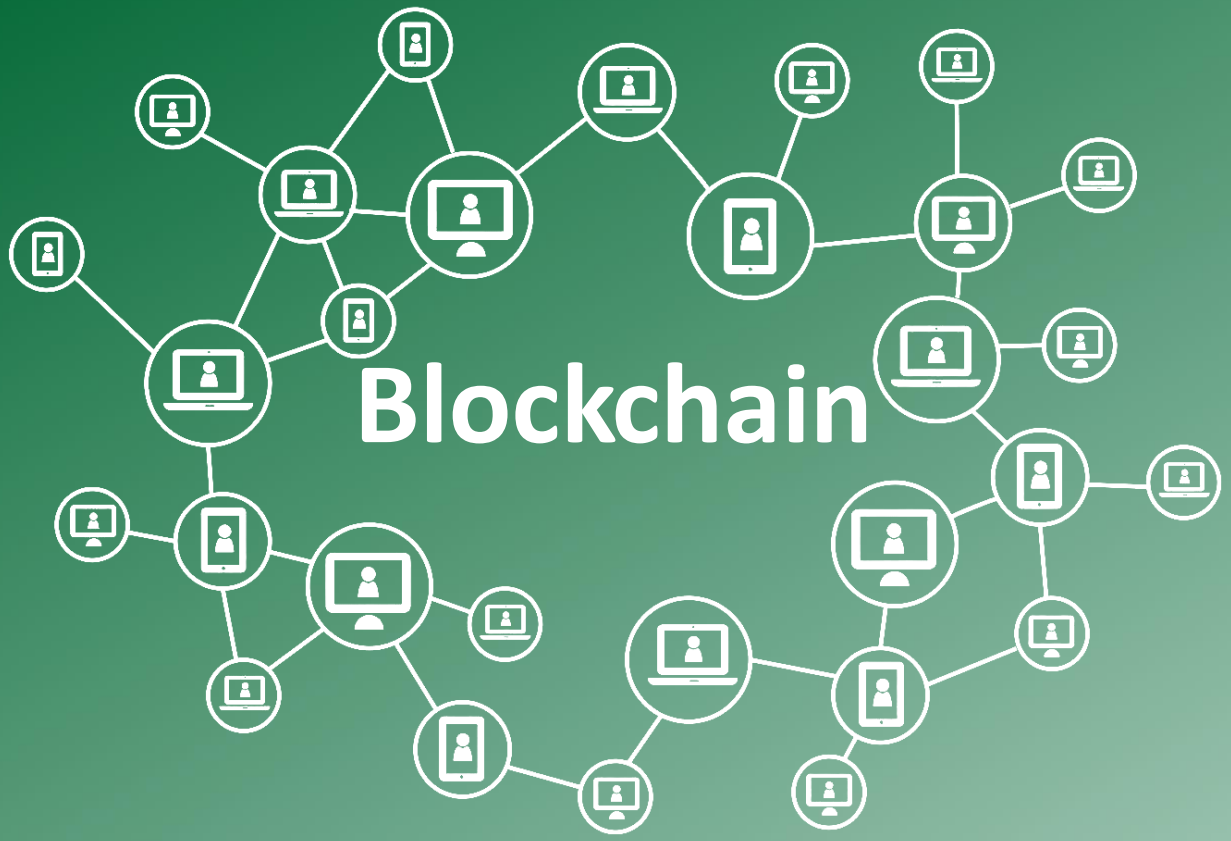
- Distributed ledgers use independent computers (referred to as nodes) to record, share and synchronize transactions in their respective electronic ledgers
- Blockchain technology can be used as DLT



Centralized Ledger



Distributed Ledger



Cryptocurrency Definition & Related Terms

Definition#1

A cryptocurrency is a standard currency which is used for the sole purpose of making or receiving payments on the blockchain. For instance, the most popular cryptocurrency is Bitcoin.

Investopedia

https://www.investopedia.com/terms/c/crypto-token.asp

Definition#2

Cryptocurrency is a form of digital money that is designed to be secure and, in many cases, anonymous. It is a currency associated with the internet that uses cryptography, the process of converting legible information into an almost uncrackable code, to track purchases and transfers.

The Telegraph

https://www.telegraph.co.uk/technology/0/cryptocurrency/

#	Name	Market Cap	Price	Volume (24h)	Circulating Supply	Change (24h)	Price Graph (7d)
1	Bitcoin	$99,595,672,864	$5,451.81	$73,337,104,284	18,268,375 BTC	-15.75%	
2	Ethereum	$14,184,309,601	$128.86	$28,942,056,755	110,077,141 ETH	-11.95%	
3	XRP	$6,727,948,121	$0.153543	$4,967,946,403	43,818,008,717 XRP *	-7.83%	
4	Tether	$4,669,079,945	$1.01	$98,467,098,612	4,642,367,414 USDT *	0.18%	
5	Bitcoin Cash	$3,060,470,620	$166.98	$7,060,269,838	18,328,138 BCH	-16.51%	
6	Litecoin	$2,194,990,986	$34.15	$7,373,370,689	64,279,418 LTC	-7.25%	
7	Bitcoin SV	$2,060,116,905	$112.42	$3,282,124,618	18,325,177 BSV	-22.09%	
8	EOS	$1,793,559,571	$1.95	$9,448,747,053	920,888,980 EOS *	-14.47%	
9	Binance Coin	$1,607,645,102	$10.34	$705,075,146	155,536,713 BNB *	-16.63%	
10	Tezos	$1,140,321,169	$1.62	$438,751,167	703,538,921 XTZ *	-8.44%	

Source: CoinMarketCap, June 20, 2018, https://coinmarketcap.com/



Source: CoinMarketCap, June 20, 2018, https://coinmarketcap.com/

Mining

- Cryptocurrency mining includes two functions:
  - Adding transactions to the blockchain (securing and verifying)
  - Releasing new currency. Individual blocks added by miners should contain a proof-of-work, or PoW.
- Mining needs a computer and a special program for miners to compete with their peers in solving complicated mathematical problems.
- The problem (for Bitcoin) is to zero in on a hash value less than the target and the first to crack it would be considered as the one who mined the block and is eligible to get a rewarded.

Mining Rig



Wallet

- A cryptocurrency wallet is a software program that stores or manage private and public keys and interacts with one or more cryptocurrencies to enable users to send and receive cryptocurrency and monitor their balance.
- Cryptocurrencies don’t get stored in any single location or exist anywhere in any physical form. All that exists are records of transactions stored on the blockchain.



Different Types of Cryptocurrency Wallets

- Desktop: Software wallet
- installed on single PC

  Online: Wallets store your private keys online and are controlled by a third party

- Mobile: Wallets run on an app on your phone

- Hardware: A hardware device like a USB to store a user's private keys

  Paper: A physical copy or
- printout of your public and private keys





Are Wallets Secure?

- The level of security depends on the type of wallet you use (desktop, mobile, online, paper, hardware) and the service provider.
- Online wallets can expose users to possible vulnerabilities in the wallet platform which can be exploited by hackers to steal your funds.
- Offline wallets, on the other hand, cannot be hacked but easy to be lost.
- Remember that no matter which wallet you use, losing your private keys will lead you to lose your money.

Double Spending

- A double spend is an attack where the given set of coins is spent in more than once. There are a couple main ways to perform a double spend:
  - Send two conflicting transactions in rapid succession into the cryptocurrency network. This is called a race attack.
  - Pre-mine one transaction into a block and spend the same coins before releasing the block to invalidate that transaction. This is called a Finney attack.
  - Own 51+% of the total computing power of the cryptocurrency network to reverse any transaction you feel like, as well as have total control of which transactions appear in blocks. This is called a 51% attack.
- To prevent damages
  - Race attack - wait for one confirmation to appear on a given transaction.
  - Finney attack - wait for 6 confirmations to appear on a transaction, or less if the transaction is small (but still require at least 1)
  - 51% attack: don’t worry

Premined Coins/Tokens

- A premine is where a developer allocates a certain amount of currency credit to a particular address before releasing the source code to the open community.
- For example, Ethereum's Ether generation
  - 60 million Ether created to contributors of the presale
  - 12 Million (20% of the above) were created to the development fund, most of it going to early contributors and developers and the remaining to the Ethereum Foundation

Bithumb $31 Million Crypto Exchange Hack: What We Know (And Don't)



Wolfie Zhao

Jun 20, 2018 at 09:10 UTC | Updated Jun 20, 2018 at 12:19 UTC

NEWS

On Wednesday, roughly 35 billion Korean won (around $31 million) in cryptocurrency was stolen by hackers from the South Korea-based exchange Bithumb.

Although the breach may not be as severe as the $530 million hack of the Coincheck exchange earlier this year, the fact that Bithumb now ranks as the sixth biggest trading venue in the world still marks it as a notable, and worrying, incident.

While more details about the heist have surfaced in the hours following the event's confirmation, providing a glimpse into Bithumb's internal operations, some important questions about the hack still remain unanswered.

Here's what we know about the hack so far, and some details we still don't.

What we know

XRP compromised **Ripple**

Source: CoinDesk

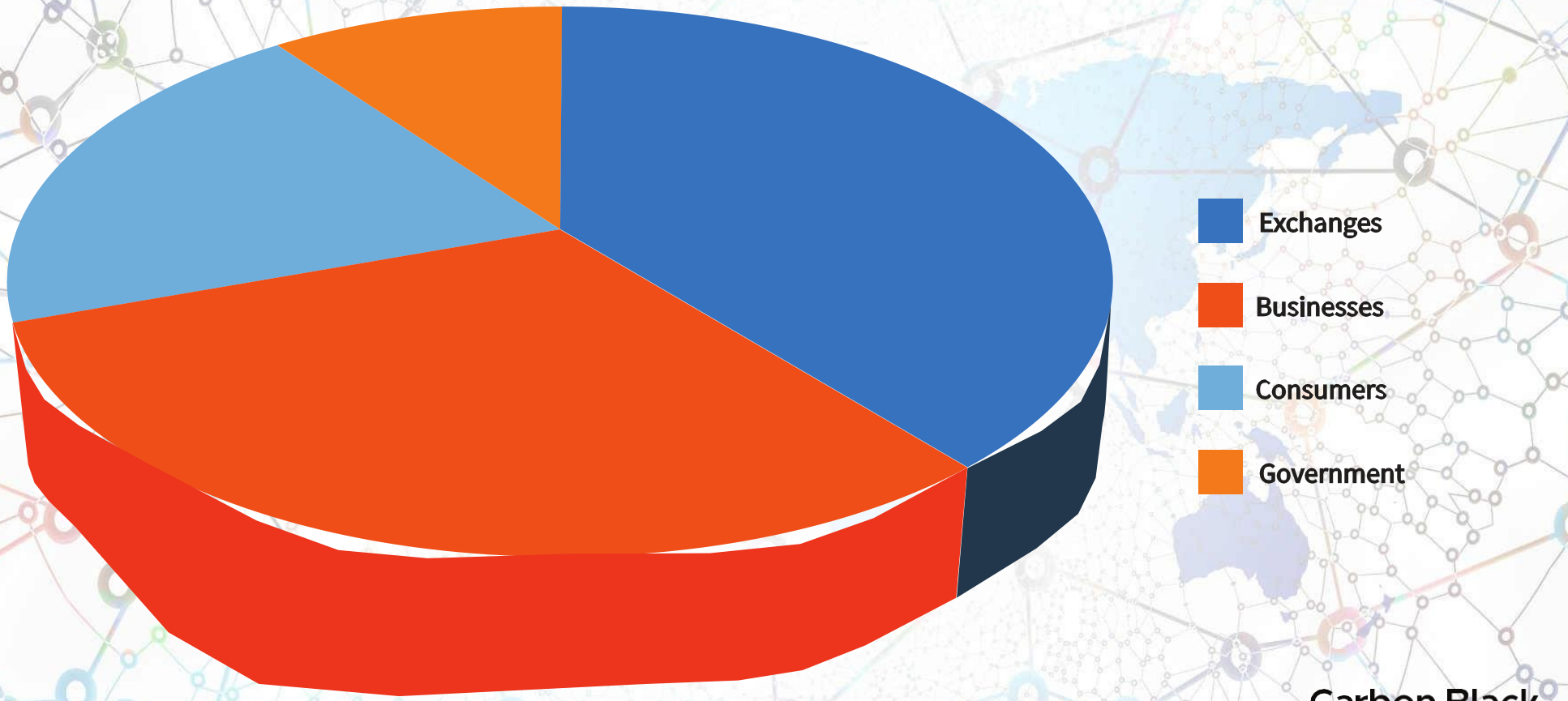
https://www.coindesk.com/bithumb-exchanges-31-million-hack-know-dont-know/

"Modern-day cybercriminals are increasingly using the dark web to facilitate cryptocurrency theft on a large scale."

Carbon Black

Source: Cryptocurrency Gold Rush on the Dark Web, Carbon Black, June 2018

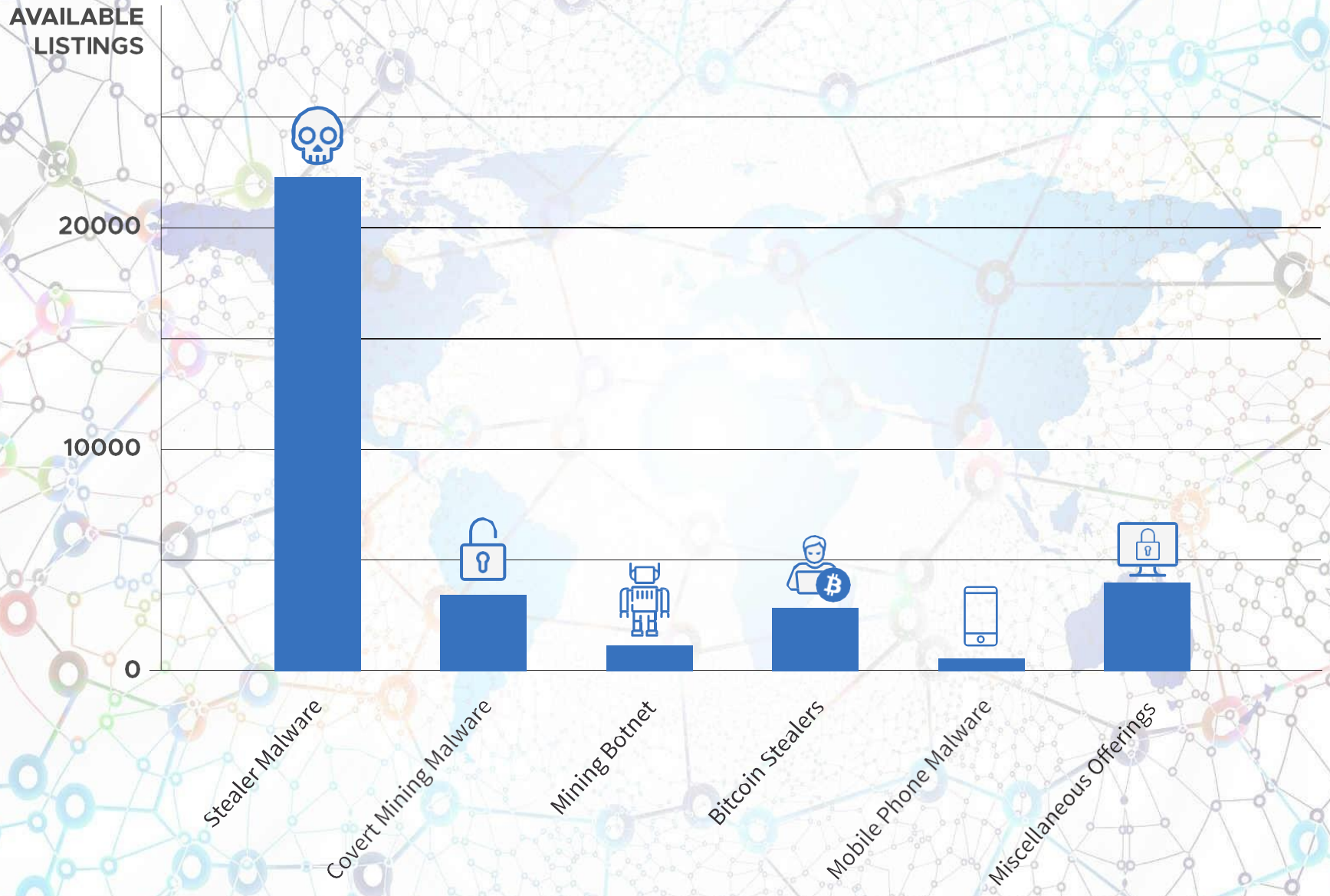
Most Often Targeted by Cryptocurrency-Related Attacks



Carbon Black.

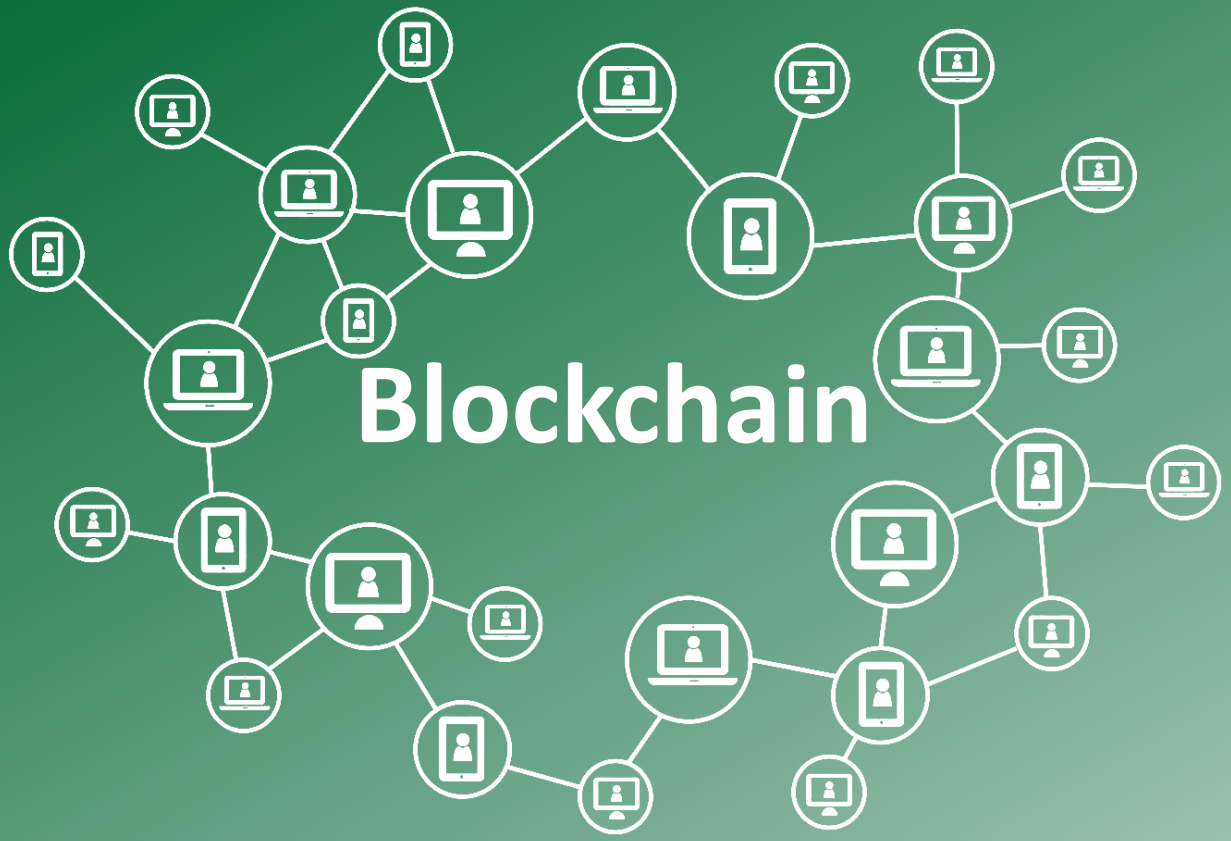
Source: Cryptocurrency Gold Rush on the Dark Web, Carbon Black, June 2018

Key Tactics, Techniques, and Procedures (TTPs) Most Favored by Cybercriminals



Carbon Black.

Source: Cryptocurrency Gold Rush on the Dark Web, Carbon Black, June 2018



Blockchain Use Cases

National Digital Identity Platform (NDID)

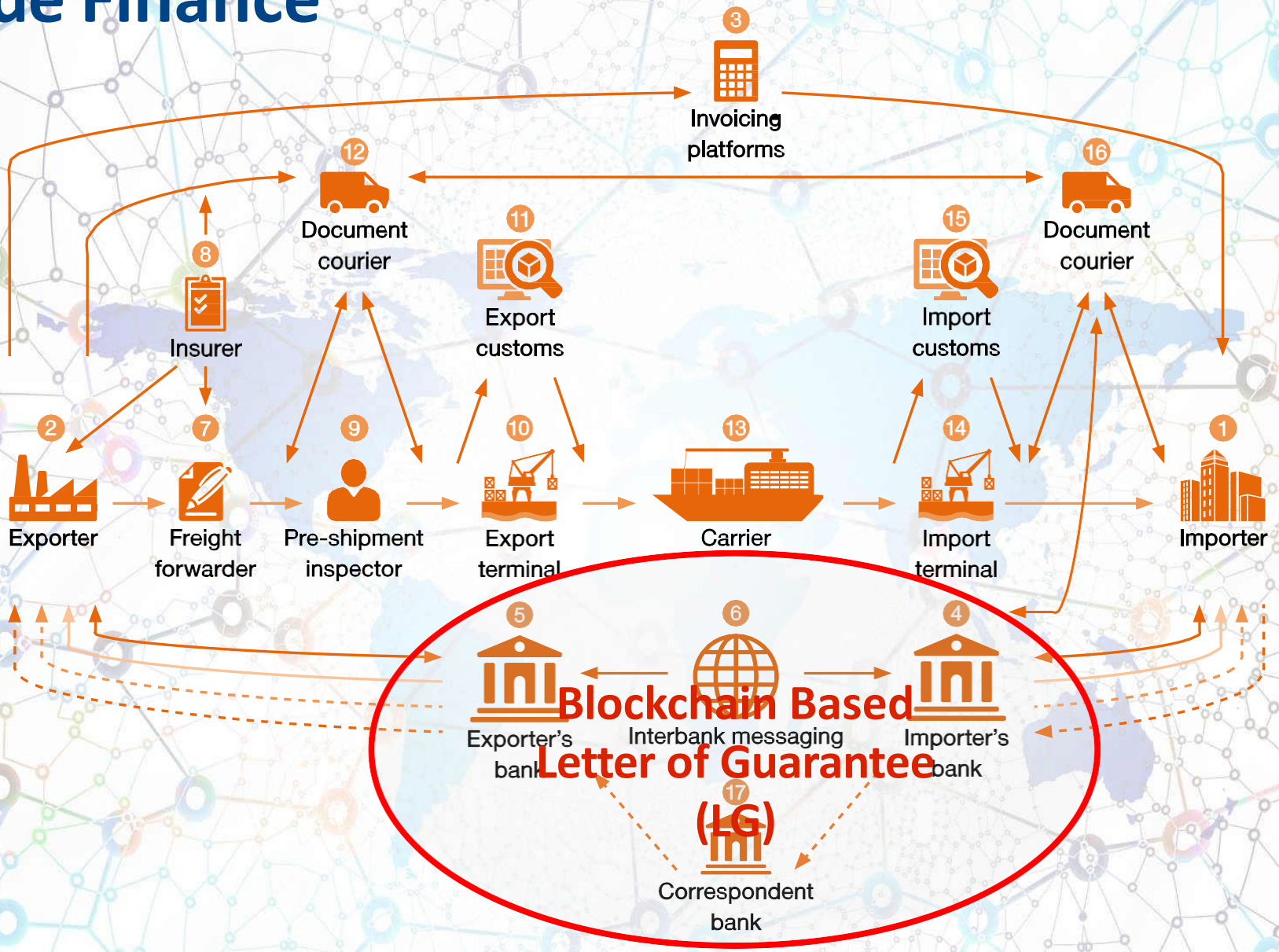
- The Digital Identity Platform is intended to provide a flexible and highly secured method of self-
- identification for any individual person and juristic person.

  Leverage any reliable identity the user currently holds. Examples of reliable identity could be, for example, Citizen ID, Bank Accounts, Passport Number, Tax ID, Biometric Data.

NDID Terms

- RP (Relying Party)
  - An entity that relies upon the subscriber's authenticator(s) and credentials or a verifier's assertion of a claimant's identity, typically to process a transaction or grant access to information or a system.
- IdP (Identity Provider)
  - An entity that creates, maintains, and manages identity information for principals while providing authentication services to relying party applications within a distributed network
- AS (Authoritative Source)
  - An entity that provides the truth of information related to each principal when that principal makes a consent.
  - An entity that has access to, or verified copies of, accurate information from an issuing source during identity proofing.

Trade Finance



Source: Use Cases for Blockchain Technology in Energy & Commodity Trading, PricewaterhouseCoopers
https://www.pwc.com/gx/en/industries/assets/blockchain-technology-in-energy.pdf


Blockc
Cryptoc
FOR DUMMIES
DUMMIES
Q&A